Sample Size and Geometry

Alfred Baron, 2023, 2025, 2026

Please also read the advice for single crystals (if you are measuring a single crystal).

Sample Size & Shape & Mounting

The optimal and minimum acceptable sample sizes depend very much on the measurement desired: How weak is the signal? What temperature range? What Q range? How much background is acceptable if the tail of the beam hits the holder? How well defined is the measurement? - do you know exactly what mode at what Q? How much beamtime is there?

Generally, bigger samples, up to a few mm, are better. The smallest samples we have measured are ~0.005 mm but comfortable size samples are typically 1~2 mm scale (thickness one attenuation length if Laue/transmission geometry). We often measure samples of transverse dimensions 0.2~0.5 mm – but note these require care to mount (mounting is the responsibility of the sample provider).

Consider the scattering from the sample holder. By default the beam will have tails out to ~0.15 mm from the beam center. If the tail of the beam hits the sample holder, or glue holding the sample, that will give background in the IXS measurement.

Platelets are preferred for IXS in a transmission/Laue geometry. Needles are usually a poor choice for IXS as they unnecessarily reduce rates and increase alignment work (e.g. see the note on cryostat chi motion below).

Flat and lightly etched or cleaved surfaces are preferred. Surface structure (imperfections, microcrystals, strain, curvature) often can lead to backgrounds or other issues.

Avoid choices that negatively impact measurements. Well set-up experiments can have relatively high rates, easy alignment, and be completely background free. To achieve this, great effort has been invested to optimize the beamline performance. Consider this carefully before making any decision regarding sample preparation/selection/mounting/etc. that might reduce that performance. Also recall that all meV IXs experiments are rate-limited.

Transmission/Laue Geometry vs Reflection/Bragg Geometry:

Please see the "xray_effective_path_length()" command in SIXCIRCLE to estimate relative count rates in different geometries

A transmission or Laue geometry is usually the most flexible allowing access to the largest range of accessible momentum transfers. It can be the best geometry if you want very low momentum transfers. It is also generally less sensitive to surface preparation/quality. The ideal thickness for transmission is 1 attenuation length. Practically between 1 and 2 is usually OK. Much less than 1 or more than 2 essentially wastes beam time. It is usually easiest if the transverse dimensions of the sample are 1 to 5 mm. See also the note about the cryostat below.

A reflection or Bragg geometry (for thick samples) generally can provide higher rates than a transmission geometry, but limits one to total momentum transfers, **Q**, that are not too far from the surface normal (n.b., little q, **q,** is *not* limited in the same way). If you know where (what **Q**) you want to measure a Bragg geometry can be a very good choice. Note, however, that spectra can be sample surface sensitive and you need a flat surface large enough to comfortably intercept the entire transverse beam size in all Bragg geometries of interest. Flat surfaces of ~2-5 mm length scale are easiest. Also, lightly etching the surface can remove strain and reduce elastic scattering in the phonon spectra. For samples with small lattice constants also consider how to access two alignment reflections (primary and secondary).

Chi & Sample Stability when using a Cryostat/Cryofurnace.

In general the spectrometer has a sphere of confusion of a few 10's of microns – sample positions when mounted in the center of the circles will tend to stay stable at that level. **However, if the cryostat or cryofurnace is mounted, these have a center of mass that is far outside the center**

of circles and when chi is changed by more than (roughly) +-25 degrees, the sample position can shift by 0.1 mm (even 0.5 mm for chi ~90 degrees). The position is motorized, and can be corrected, but this needs to be done (usually by hand) and can slow an experiment. This can slow the experiment a lot if you need to work at larger chi-values and change them often.

Determining grazing angles onto the sample surface.

Diffractometer codes usually can provide the x-ray beam grazing angles onto and off of the sample surface. **For spec and related codes, set the sample outward-pointing normal vector (in H,K,L) using the "setaz" command and then alpha is the angle onto the sample surface and beta is the angle out of the sample surface into the center of the tth arm - analyzer 6** at BL43LXU. If alpha and beta are the same sign then you have a Bragg/reflection geometry, if they are different signs, a Laue/transmission geometry. Note that having alpha small (say <~5 deg) can make alignment to the incident beam tricky while having beta small (<~5 deg) can mean some analyzers may be blocked by the sample surface.